

105th FoCARS

Foundation Course For Agricultural Research Service

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Human Resources Management

- Overview of HRM in Indian NARS
- Time Management
- Interpersonal Behaviour & Relationships
- Leadership styles
- Motivation
- Stress: Assessment and Management
- Group Dynamics & Teamwork
- Transactional Analysis
- **Creativity and Problem Solving**
- Emotional Intelligence
- Groups in Organization
- Presentation Skills
- Ethics in Agricultural Research

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CREATIVITY AND PROBLEM SOLVING

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What is Creativity?

An Ability. A simple definition is that creativity is the ability to imagine or invent something new. Creativity is not the ability to create out of nothing, but the ability to generate new ideas by combining, changing, or reapplying existing ideas. Some creative ideas are astonishing and brilliant, while others are just simple, good, practical ideas that no one seems to have thought of yet.

Believe it or not, everyone has substantial creative ability. Just look at how creative children are. In adults, creativity has too often been suppressed through education, but it is still there and can be reawakened. Often all that's needed to be creative is to make a commitment to creativity and to take the time for it.

An Attitude. Creativity is also an attitude: the ability to accept change and newness, a willingness to play with ideas and possibilities, a flexibility of outlook, the habit of enjoying the good, while looking for ways to improve it. We are socialized into accepting only a small number of permitted or normal things, like chocolate-covered strawberries, for example. The creative person realizes that there are other possibilities, like peanut butter and banana sandwiches, or chocolate-covered prunes.

A Process. Creative people work hard and continually to improve ideas and solutions, by making gradual alterations and refinements to their works. Contrary to the mythology surrounding creativity, very, very few works of creative excellence are produced with a single stroke of brilliance

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or in a frenzy of rapid activity. Much closer to the real truth are the stories of companies who had to take the invention away from the inventor in order to market it because the inventor would have kept on tweaking it and fiddling with it, always trying to make it a little better.

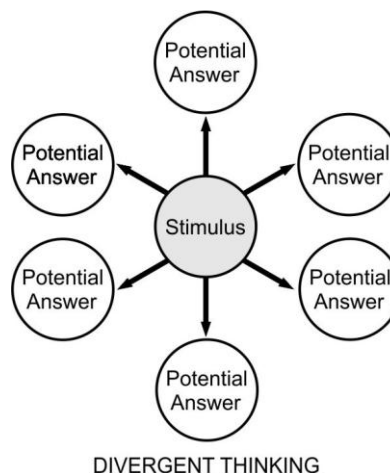
Much of the thinking done in formal education emphasizes the skills of analysis--teaching students how to understand claims, follow or create a logical argument, figure out the answer, eliminate the incorrect paths and focus on the correct one. However, there is another kind of thinking, one that focuses on exploring ideas, generating possibilities, looking for many right answers rather than just one. Both of these kinds of thinking are vital to a successful working life, yet the latter one tends to be ignored until after college. We might differentiate these two kinds of thinking like this:

Critical Thinking	Creative Thinking
analytic	generative
convergent	divergent
vertical	lateral
probability	possibility
judgment	suspended judgment
focused	diffuse
objective	subjective
answer	an answer
left brain	right brain
verbal	visual
linear	associative
reasoning	richness, novelty
yes but	yes and

Divergent Thinking

Divergent thinking is a thought process or method used to generate creative ideas by exploring many possible solutions. It is often used in conjunction with its cognitive colleague, convergent thinking, which follows a particular set of logical steps to arrive at one solution, which in

some cases is a 'correct' solution. By contrast, divergent thinking typically occurs in a spontaneous, free-flowing, 'non-linear' manner, such that many ideas are generated in an emergent cognitive FASHION. Many possible solutions are explored in a SHORT amount of time, and unexpected connections are drawn. After the process of divergent thinking has been completed, ideas and information are organized and structured using convergent thinking. The psychologist J.P. **Guilford** first coined the terms convergent thinking and divergent thinking in 1959.



Elements of Divergent Thinking

1. **Complexity** – The capacity to conceptualize difficult, multifaceted, many layered or intricate products or ideas.
2. **Curiosity** – The personality characteristic of displaying probing behaviors, searching, asking questions, learning to get more knowledge/information about something, and of being able to go deeper into ideas.
3. **Elaboration** – The skill of adding to, building off of or embellishing a product or an idea.
4. **Flexibility** – The capability of creating varied perceptions or categories wherefrom come a range of different ideas pertaining to the same thing or problem.

5. **Fluency** – The skill of engendering many ideas so as to have an increase in the number of potential solutions or associated products.
6. **Imagination** – The capability of dreaming up, inventing, or to think, to see, to conceptualize novel products or ideas, to be original.
7. **Originality** – The skill of coming up with fresh, unusual, unique, extremely different or completely new products or ideas.
8. **Risk-taking** – The readiness to be courageous, daring, adventuresome – take risks or experiment with new things so as to stand apart.

The Creative Process

Creativity involves creative process. The creative process involves creative thinking. Creative thinking involves some specific and definite dimensions or steps or stages. **Wallas** (1926) has emphasized that there are four stages, namely preparation, incubation, inspiration or illumination and verification or revision. At later period, implementation is also added as one of the stages of creative process.

1. Preparation

During the preparation step of the creative process model, an individual becomes curious after encountering a problem. Examples of problems can include an artistic challenge or an assignment to write a paper. During this stage, she may perform research, creates goals, organize thoughts and brainstorm as different ideas formulate. For example, a marketing professional may prepare for a marketing campaign by conducting market research and formulating different advertisement ideas.

2. Incubation

While the individual begins to process her ideas, she begins to synthesize them using her imagination and begins to construct a creation. During this step, the individual does not actively try to find a solution, but continues to mull over the idea in the back of her head.

3. Inspiration or Illumination

As ideas begin to mature, the individual has an epiphany regarding how to piece her thoughts together in a manner that makes sense. The moment of illumination can happen unexpectedly. For example, an individual with the task of putting together an office party may have an idea for a theme while driving home from work.

4. Verification or Revision

After a solution reveals itself in an epiphany, the individual then evaluates whether the insight is worth the pursuit. He may make changes to his solution so it is clearer. He may consult with peers or supervisors regarding his insights during this step before pursuing it further. If he works with clients, he may seek a client's input and approval before moving on to the next step.

5. Implementation

The implementation of an idea or solution in the creative process model is when an individual begins the process of transforming her thoughts into a final product. For example, during this step, a painter may begin outlining shapes on a canvas with charcoal before applying oil paints to the medium. An individual may begin this step more than once in order to reach the desired outcome. For example, a graphic designer may open a new digital canvas if he did not have the scale calculated correctly on a previous work, and he will continue to implement his ideas and make adjustments until he reaches a pleasing final product.

Characteristics of the Creative Person

- curious
- seeks problems
- enjoys challenge
- optimistic
- able to suspend judgment
- comfortable with imagination
- sees problems as opportunities
- sees problems as interesting

- problems are emotionally acceptable
- challenges assumptions
- doesn't give up easily: perseveres, works hard

Negative Attitudes That Block Creativity

1. Oh no, a problem! The reaction to a problem is often a bigger problem than the problem itself. Many people avoid or deny problems until it's too late, largely because these people have never learned the appropriate emotional, psychological, and practical responses. A problem is an opportunity. The happiest people welcome and even seek out problems, meeting them as challenges and opportunities to improve things. Definition: a problem is (1) seeing the difference between what you have and what you want or (2) recognizing or believing that there is something better than the current situation or (3) an opportunity for a positive act. Seeking problems aggressively will build confidence, increase happiness, and give you a better sense of control over your life.

2. It can't be done. This attitude is, in effect, surrendering before the battle. By assuming that something cannot be done or a problem cannot be solved, a person gives the problem a power or strength it didn't have before. And giving up before starting is, of course, self-fulfilling. But look at the history of solutions and the accompanying skeptics: man will never fly, diseases will never be conquered, rockets will never leave the atmosphere. Again, the appropriate attitude is summed up by the statement, "The difficult we do immediately; the impossible takes a little longer."

3. I can't do it. Or There's nothing I can do. Some people think, well maybe the problem can be solved by some expert, but not by me because I'm not (a) smart enough, (b) an engineer, or (c) a blank (whether educated, expert, etc.) Again, though, look at the history of problem solving.

Who were the Wright brothers that they could invent an airplane? Aviation engineers? No, they were bicycle mechanics. The ball point pen was invented by a printer proofreader, Ladislao Biro, not a mechanical engineer.

Major advances in submarine design were made by English clergyman G. W. Garrett and by Irish schoolmaster John P. Holland. The cotton gin was invented by that well known attorney and tutor, Eli Whitney. The fire extinguisher was invented by a captain of militia, George Manby.

And so on. In fact, a major point made by recent writers about corporate excellence is that innovations in industry almost always come from individuals (not research groups) outside of the area of the invention. General Motors invented Freon, the refrigeration chemical, and tetraethyl lead, the gasoline additive. Kodachrome was invented by two musicians. The continuous steel casting process was invented by a watchmaker (fooling around with brass casting). Soap making chemists turned down the problem of inventing synthetic detergents: those detergents were invented by dye making chemists.

In a nutshell, a good mind with a positive attitude and some good problem solving skills will go far in solving any problem. Interest in and commitment to the problem are the keys. Motivation--a willingness to expend the effort--is more important than laboratory apparatus. And remember that you can always do something. Even if you cannot totally eradicate the problem from the face of the earth, you can always do something to make the situation better.

4. But I'm not creative. Everyone is creative to some extent. Most people are capable of very high levels of creativity; just look at young children when they play and imagine. The problem is that this creativity has been suppressed by education. All you need to do is let it come back to the surface. You will soon discover that you are surprisingly creative.

5. That's childish. In our effort to appear always mature and sophisticated, we often ridicule the creative, playful attitudes that marked our younger years. But if you solve a problem that saves your marriage or gets you promoted or keeps your friend from suicide, do you care whether other people describe your route to the solution as "childish?" Besides, isn't play a lot of fun? Remember that sometimes people laugh when

something is actually funny, but often they laugh when they lack the imagination to understand the situation.

6. What will people think? There is strong social pressure to conform and to be ordinary and not creative. The constant emphasis we see in society is toward the ruthlessly practical and conformist. Even the wild fashions, from those in Vogue to punk rock, are narrowly defined, and to deviate from them is considered wrong or ridiculous. Some peoples' herd instinct is so strong that they make sheep look like radical individualists.

So, what will people think? Well, they're already talking about you, saying that your nose is too big or your shoes are funny or you date weird people. So, since others are going to talk about you in unflattering ways anyway, you might as well relax and let your creativity and individualism flow.

Almost every famous contributor to the betterment of civilization was ridiculed and sometimes even jailed. Think about Galileo. And look what happened to Jesus. *"Progress is made only by those who are strong enough to endure being laughed at."* Solutions are often new ideas, and new ideas, being strange, are usually greeted with laughter, contempt, or both. That's just a fact of life, so make up your mind not to let it bother you. Ridicule should be viewed as a badge of real innovative thinking.

7. I might fail. Thomas Edison, in his search for the perfect filament for the incandescent lamp, tried anything he could think of, including whiskers from a friend's beard. In all, he tried about 1800 things. After about 1000 attempts, someone asked him if he was frustrated at his lack of success. He said something like, "I've gained a lot of knowledge--I now know a thousand things that won't work."

Fear of failure is one of the major obstacles to creativity and problem solving. The cure is to change your attitude about failure. Failures along the way should be expected and accepted; they are simply learning tools that help focus the way toward success. Not only is there nothing wrong with failing, but failing is a sign of action and struggle and attempt--much better than inaction. The go-with-the- flow types may never fail, but they

are essentially useless to humanity, nor can they ever enjoy the feeling of accomplishment that comes after a long struggle.

Suppose you let your fear of failure guides your risk taking and your attempts. You try only three things in a year because you are sure of succeeding. At the end of the year the score is: Successes 3, Failures 0. Now suppose the next year you don't worry about failing, so you try a hundred things. You fail at 70 of them. At the end of the year the score is Successes 30, Failures 70. Which would you rather have--three successes or 30--ten times as many? And imagine what 70 failures will have taught you. Proverb: *Mistakes aren't fun, but they sure are educational.*

Problem Solving

Meaning and Definition

From birth onwards, everybody in this world is beset with some problem or the other. There are needs and motives that are to be satisfied. For this purpose, definite goals or aims are set. In an attempt for their realization one experiences obstacles and interferences in one's attempt to achieve them. This create problems and serious and deliberate efforts have to be made to overcome these impediments.

The productive work involved in the evaluation of the situation and the strategy worked out to reach one's goals is collectively termed as problem solving. This is an essential exercise for individual advancement as also for the advancement of society. The meaning and nature of problem solving is further clarified by the following definitions:

Problem-solving behaviour occurs in novel or difficult situations in which a solution is not obtained by the habitual methods of applying concepts and principles derived from past experiences in very similar situations (Woodworth and Marquis, 1948).

Problem-solving is a process of overcoming difficulties that appear to interfere with the attainment of a goal. It is a procedure of making adjustment in spite of interferences (Skinner, 1968).

An analysis of the above definitions brings out the following observations about the meaning and nature of problem-solving behaviour:

1. In the satisfaction of one's needs and realization of the set goals, problem-solving behaviour arises only when the goal is purposeful and essential for the individual. There is a serious interference in the realization of this goal, and this interference or obstacle cannot be overcome by simple habitual acts or mechanical trial and error methods.
2. One has to utilize one's thinking and reasoning powers and engage in serious mental work by systematically following some well-organized steps for the removal of the difficulties and obstacles.
3. The problem-solving behaviour involves quite deliberate, conscious and serious effort on the part of the problem-solver.
4. Problem-solving behaviour helps in the removal of, or adjustment with, interferences and ultimately helps an individual to reach his goal and satisfy his motives.
5. Problem-solving behaviour helps an individual in the growth and development of his personality, making his life happier and wiser by appropriate adjustment. It also contributes significantly to the progress and development of society.

In view of the foregoing, problem-solving behaviour may be said to be a deliberate and purposeful act on the part of an individual to realize the set goals or objectives by inventing some novel methods or systematically following some planned steps for removal of the interferences and obstacles in the path of the realization of these goals when usual methods like trial and error, habit formation and conditioning fail.

Steps in Effective Problem-Solving Behaviour

Psychologists have tried to study carefully, the behaviour involved in the process of problem-solving in animals as well as human beings. They have suggested different steps involved in the process of problem-solving according to their respective findings and viewpoints.

Bransford and Barry Stein (1984) advocated five steps that are basically associated with the task of problem-solving. They referred to these steps as **IDEAL** thinking and arranged them in the following order:

I = Identifying the problem

D = defining and representing the problem

E = Exploring possible strategies

A = Acting on the strategies

L = Looking back and evaluating the effect of one's activities

Bourne, Dominowski and Loftus (1979), on the other hand, enumerated three steps or stages in problem-solving: preparation, production and evaluation, by proclaiming "*we prepare, we produce and we evaluate in the task of problem-solving*".

Problem-solving is an individual phenomena and involves the exercise of cognitive abilities of a higher order and continuous and persistent struggling on the conscious as well as unconscious levels. Often, there is a considerable movement back and forth as one moves from one step to another in the task of problem-solving.

Myths about Creative Thinking and Problem Solving

1. Every problem has only one solution (or one right answer). The goal of problem solving is to solve the problem, and most problems can be solved in any number of ways. If you discover a solution that works, it is a good solution. There may be other solutions thought of by other people, but that doesn't make your solution wrong. What is THE solution to putting words on paper? Fountain pen, ball point, pencil, marker, typewriter, printer, Xerox machine, printing press?

2. The best answer/solution/method has already been found. Look at the history of any solution set and you'll see that improvements, new solutions, new right answers, are always being found. What is the solution to human transportation? The ox or horse, the cart, the wagon, the train, the car, the airplane, the jet, the SST? Is that the best and last? What about pneumatic tubes, hovercraft, even Star trek type beams?

What is the best way to put words on paper? The word processor? Is that the last invention? How about voice recognition, or thought wave input?

On a more everyday level, many solutions now seen as best or at least entrenched were put in place hastily and without much thought--such as the use of drivers' licenses for ID cards or social security numbers for taxpayer ID numbers. Other solutions are entrenched simply for historical reasons: they've always been done that way. Why do shoe laces still exist, when technology has produced several other, better ways to attach shoes to feet (like velcro, elastic, snap buttons, and so on)?

3. Creative answers are complex technologically. Only a few problems require complex technological solutions. Most problems you'll meet with require only a thoughtful solution requiring personal action and perhaps a few simple tools. Even many problems that seem to require a technological solution can be addressed in other ways.

For example, what is the solution to the large percentage of packages ruined by the Post Office? Look at the Post Office package handling method. Packages are tossed in bins when you send them. For the solution, look at United Parcel. When you send a package, it is put on a shelf. The change from bin to shelf is not a complex or technological solution; it's just a good idea, using commonly available materials.

As another example, when hot dogs were first invented, they were served to customers with gloves to hold them. Unfortunately, the customers kept walking off with the gloves. The solution was not at all complex: serve the hot dog on a roll so that the customer's fingers were still insulated from the heat. The roll could be eaten along with the dog. No more worries about disappearing gloves. (Note by the way what a good example of

changing direction this is. Instead of asking, "How can I keep the gloves from being taken?" the hot dog server stopped thinking about gloves altogether.)

4. Ideas either come or they don't. Nothing will help. There are many successful techniques for stimulating idea generation. We will be discussing and applying them.

Mental Blocks to Creative Thinking and Problem Solving

1. Prejudice. The older we get, the more preconceived ideas we have about things. These preconceptions often prevent us from seeing beyond what we already know or believe to be possible. They inhibit us from accepting change and progress.

2. Functional fixation. Sometimes we begin to see an object only in terms of its name rather than in terms of what it can do. Thus, we see a mop only as a device for cleaning a floor, and do not think that it might be useful for clearing cobwebs from the ceiling, washing the car, doing aerobic exercise, propping a door open or closed, and so on.

There is also a functional fixation of businesses. In the late nineteenth and early twentieth centuries the railroads saw themselves as railroads. When automobiles and later airplanes began to come in, the railroads didn't adapt. "That's not our business," they said. But if they had seen themselves as in the people transportation business rather than in the railroad business, they could have capitalized on a great opportunity.

And there's a functional fixation of people, too. Think a minute how you react when you see your pastor mowing his lawn, or your auto mechanic on a television show promoting a book. Stereotyping can even be a form of functional fixation--how many people would laugh at a blonde quoting Aristotle? Too often we permit only a narrow range of attitudes and behaviors in other people, based on bias, prejudice, hasty generalization, or limited past experience. Think of those statements like, "I can't believe he said that," or "Imagine her doing that," and so on. But recall the proverb, "The goal of my life is not to live down to your expectations."

3. Learned helplessness. This is the feeling that you don't have the tools, knowledge, materials, ability, to do anything, so you might as well not try. We are trained to rely on other people for almost everything. We think small and limit ourselves. But the world can be interacted with.

If you are in need of information, there are libraries, bookstores, friends, professors, and, of course, the Internet. And there are also city, county, and state government agencies with addresses and phone numbers and web sites. There are thousands of government agencies that really exist and that will talk to you. Contact the EPA if you're working on air pollution or pesticides. Get some government publications. Call your state senator or federal congressman for help on bills, information, problems. Contact the manufacturer of a product to find out what you want to know about it.

If you are technologically poor, you can learn. Learn how to cook, use tools, make clothes, use a computer. You can learn to do anything you really want to do. All you need is the motivation and commitment. You can learn to fly an airplane, drive a truck, scuba dive, fix a car--name it.

4. Psychological blocks. Some solutions are not considered or are rejected simply because our reaction to them is "Yuck." But icky solutions themselves may be useful or good if they solve a problem well or save your life. Eating lizards and grasshoppers doesn't sound great, but if it keeps you alive in the wilderness, it's a good solution.

Perhaps more importantly, what at first seem to be icky ideas may lead to better solutions--de-ickified analogues of the original. When doctors noted that some unsophisticated natives were using giant ant heads to suture wounds, they imitated this pincer-closing technique by inventing the surgical staple. Psychological blocks prevent you from doing something just because it doesn't sound good or right, which is a pretty ridiculous thing. Overcoming such blocks can be really beneficial.

Positive Attitudes for Creativity

1. Curiosity. Creative people want to know things--all kinds of things--just to know them. Knowledge does not require a reason. The question,

"Why do you want to know that?" seems strange to the creative person, who is likely to respond, "Because I don't know the answer." Knowledge is enjoyable and often useful in strange and unexpected ways.

For example, I was once attempting to repair something, without apparent success, when an onlooker asked testily, "Do you know what you're doing?" I replied calmly, "No, that's why I'm doing it."

Next, knowledge, and especially wide ranging knowledge, is necessary for creativity to flourish to its fullest. Much creativity arises from variations of a known or combinations of two knowns. The best ideas flow from a well-equipped mind. Nothing can come from nothing.

In addition to knowing, creative people want to know why. What are the reasons behind decisions, problems, solutions, events, facts, and so forth? Why this way and not another? And why not try this or that?

The curious person's questioning attitude toward life is a positive one, not a destructive one reflecting skepticism or negativism. It often seems threatening because too often there is no good reason behind many of the things that are taken for granted--there is no "why" behind the status quo.

2. Challenge. Curious people like to identify and challenge the assumptions behind ideas, proposals, problems, beliefs, and statements. Many assumptions, of course, turn out to be quite necessary and solid, but many others have been assumed unnecessarily, and in breaking out of those assumptions often comes a new idea, a new path, a new solution.

For example, when we think of a college, we traditionally think of a physical campus with classrooms, a library, and some nice trees. But why must college be a place (with congregated students and faculty) at all? Thus, the electronic college now exists, where students "go" to college right at home, online. Correspondence courses have existed for years, too, beginning with the challenging of the school-as-centralized-place idea.

When we think of an electric motor, we *automatically* think of a rotating shaft machine. But why assume that? Why can't an electric motor have a

linear output, moving in a straight line rather than a circle? With such a challenged assumption came the linear motor, able to power trains, elevators, slide locks, and so on.

3. Constructive discontent. This is not a whining, griping kind of discontent, but the ability to see a need for improvement and to propose a method of making that improvement. Constructive discontent is a positive, enthusiastic discontent, reflecting the thought, "Hey, I know a way to make that better."

Constructive discontent is necessary for a creative problem solver, for if you are happy with everything the way it is, you won't want to change anything. Only when you become discontent with something, when you see a problem, will you want to solve the problem and improve the situation.

One of the hallmarks of the constructively discontented person is that of a problem seeking outlook. The more problems you find, the more solutions and therefore improvements you can make. Even previously solved problems can often be solved again, in a better way. A constructively discontent person might think, "This is an excellent solution, but I wonder if there isn't another solution that works even better (or costs less, etc)."

Another mark of constructive discontent is the enjoyment of challenge. Creative people are eager to test their own limits and the limits of problems, willing to work hard, to persevere and not give up easily. Sometimes the discontent is almost artificial--they aren't really unhappy with the status quo of some area, but they want to find something better just for the challenge of it and the opportunity to improve their own lives and those of others.

4. A belief that most problems can be solved. By faith at first and by experience later on, the creative thinker believes that something can always be done to eliminate or help alleviate almost every problem. Problems are solved by a commitment of time and energy, and where this commitment is present, few things are impossible.

The belief in the solvability of problems is especially useful early on in attacking any problem, because many problems at first seem utterly impossible and scare off the fainter hearted. Those who take on the problem with confidence will be the ones most likely to think through or around the impossibility of the problem.

5. The ability to suspend judgment and criticism. Many new ideas, because they are new and unfamiliar, seem strange, odd, bizarre, even repulsive. Only later do they become "obviously" great. Other ideas, in their original incarnations, are indeed weird, but they lead to practical, beautiful, elegant things. Thus, it is important for the creative thinker to be able to suspend judgment when new ideas are arriving, to have an optimistic attitude toward ideas in general, and to avoid condemning them with the typical kinds of negative responses like, "That will never work; that's no good; what an idiotic idea; that's impossible," and so forth. Hospital sterilization and antiseptic procedures, television, radio, the Xerox machine, and stainless steel all met with ho-hums and even hostile rejection before their persevering inventors finally sold someone on the ideas.

Some of our everyday tools that we now love and use daily, were opposed when they were originally presented: Aluminum cookware? No one wants that. Teflon pans? They'll never sell. Erasers on pencils? That would only encourage carelessness. Computers, there's no market for more than a few, so why build them?

Remember then that (1) an idea may begin to look good only after it becomes a bit more familiar or is seen in a slightly different context or circumstance and (2) even a very wild idea can serve as a stepping stone to a practical, efficient idea. By too quickly bringing your judgment into play, these fragile early ideas and their source can be destroyed. The first rule of brainstorming is to suspend judgment so that your idea-generating powers will be free to create without the restraint of fear or criticism. You can always go back later and examine--as critically as you want--what you have thought of.

Proverb: "A crank is a genius whose idea hasn't yet caught on."

6. Seeing the good in the bad. Creative thinkers, when faced with poor solutions, don't cast them away. Instead, they ask, "What's good about it?" because there may be something useful even in the worst ideas. And however little that good may be, it might be turned to good effect or made greater.

We easily fall into either/or thinking and believe that a bad solution is bad through and through, in every aspect, when in fact, it may have some good parts we can borrow and use on a good solution, or it may do inappropriately something that's worth doing appropriately. And often, the bad solution has just one really glaring bad part, that when remedied, leaves quite a good solution.

7. Problems lead to improvements. The attitude of constructive discontent searches for problems and possible areas of improvement, but many times problems arrive on their own. But such unexpected and perhaps unwanted problems are not necessarily bad, because they often permit solutions that leave the world better than before the problem arose.

8. A problem can also be a solution. A fact that one person describes as a problem can sometimes be a solution for someone else. Above we noted that creative thinkers can find good ideas in bad solutions. Creative thinkers also look at problems and ask, "Is there something good about this problem?"

For example, soon after the advent of cyanoacrylate adhesives (super glue), it was noted that if you weren't careful, you could glue your fingers together with it. This problem--a permanent skin bond--was soon seen as a solution, also. Surgeons in Viet Nam began to use super glue to glue wounds together.

Another example, also involving glue: 3M chemists were experimenting with adhesives and accidentally came up with one that was so weak you could peel it right back off. Hold strength, shear strength, all were way below the minimum standards for any self-respecting adhesive. A glue that

won't hold? Quite a problem. But this problem was also a solution, as you now see in Post-It Notes.

9. Problems are interesting and emotionally acceptable. Many people confront every problem with a shudder and a turn of the head. They don't even want to admit that a problem exists--with their car, their spouse, their child, their job, their house, whatever. As a result, often the problem persists and drives them crazy or rises to a crisis and drives them crazy.

Creative people see problems as interesting challenges worth tackling. Problems are not fearful beasts to be feared or loathed; they are worthy opponents to be jousting with and unhorsed. Problem solving is fun, educational, rewarding, ego building, helpful to society.

Miscellaneous Good Attitudes

1. Perseverance. Most people fail because they spend only nine minutes on a problem that requires ten minutes to solve. Creativity and problem solving are hard work and require fierce application of time and energy. There is no quick and easy secret. You need knowledge gained by study and research and you must put your knowledge to work by hard thinking and protracted experimentation. You've surely read of the difficulties and setbacks faced by most of the famous inventors--how many filaments Edison tried before he found a working one, how many aircraft designs failed in the attempt to break the sound barrier. But planning to persevere is planning to succeed.

2. A flexible imagination. Creative people are comfortable with imagination and with thinking so-called weird, wild, or unthinkable thoughts, just for the sake of stimulation. During brainstorming or just mental playfulness, all kinds of strange thoughts and ideas can be entertained. And the mind, pragmatist that it is, will probably find something useful in it all. We will look at several examples of this later on.

3. A belief that mistakes are welcome. Modern society has for some reason conceived the idea that the only unforgivable thing is to fail or make a mistake. Actually failure is an opportunity; mistakes show that

something is being done. So creative people have come to realize and accept emotionally that making mistakes is no negative biggie. One chief executive of a big American corporation warns all his newly hired managers, "Make sure you make a reasonable number of mistakes." Mistakes are educational and can lead to success--because they mean you are doing something.

Nurturing and Stimulation of Creativity and Problem-Solving

Sensitiveness, self-confidence, persistence, ability to see relationship and make associations, divergent thinking, ideation, ideational fluency, flexibility and originality are some of the important creative abilities which are attached to creative output. The teachers and parents should realize the need to provide proper environment for the development of these creative abilities in children. The following suggestions are given for fostering and stimulation of creativity and problem-solving in children:

1. Freedom to express or respond
2. Opportunity for ego involvement
3. Encouraging originality and flexibility
4. Removal of hesitation and fear
5. Providing appropriate opportunities and atmosphere for creative expression
6. Developing healthy habits among children
7. Using the creative resources of the community
8. Avoidance of blocks to creative thinking
9. Proper organization of the curriculum
10. Reform in the evaluation system
11. Use of techniques for fostering creativity:
 - a. Brainstorming
 - b. Use of teaching models
 - c. Use of gaming techniques
 - d. Teaching by example

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